

LLaDa Instruct Experiments

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DeltaQuant Evaluation

LLaDA-8B-Instruct accuracy under different quantization settings. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH500	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W16A8	ActQuant	78.77	35.33	57.05	$\uparrow 0.34\%$	38.41	36.20	37.31	$\downarrow 6.07\%$
		DeltaQuant	78.62	34.49	56.56	$\downarrow 0.53\%$	39.63	35.60	37.62	$\downarrow 5.29\%$

Dataset	# fewshots	generation length	diffusion steps	block size
GSM8K	4	256	256	32
Math	4	256	256	32
HumanEval	0	256	256	32
MBPP	3	256	256	32

DeltaQuaRot Evaluation (Single Anchor)

LLaDA-8B-Instruct accuracy under QuaRot rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W4A8	QuaRot	71.65	–	–	–	35.37	36.00	35.68	↓10.16%
		DeltaQuaRot	71.80	–	–	–	32.93	36.60	34.76	↓12.47%
	W4A4	QuaRot	67.55	–	–	–	26.83	32.00	29.41	↓25.93%
		DeltaQuaRot	49.36	–	–	–	21.34	31.00	26.17	↓34.11%

DeltaDuQuant Evaluation (Single Anchor)

LLaDA-8B-Instruct accuracy under DuQuant rotation quantization setting. $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

Model	Setting	Method	Math				Code			
			GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W4A8	DuQuant	77.63	–	–	–	39.02	34.60	36.81	↓7.31%
		DeltaDuQuant	77.33	–	–	–	39.63	34.80	37.22	↓6.29%
	W4A4	DuQuant	75.74	–	–	–	35.37	29.80	32.58	↓17.96%
		DeltaDuQuant	45.49	–	–	–	19.51	23.60	21.56	↓45.73%

Uniform Reanchoring for Instruct Model

LLaDA-8B-Instruct accuracy under delta-quant on top of DuQuant/QuaRot rotations, swept over the number of full-precision anchors spread uniformly across the generation trajectory.

Model	Setting	Method	Anchors	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	—	78.85	34.87	56.86	—	39.63	39.80	39.72	—
			2	71.19	—	—	—	32.93	36.60	34.76	$\downarrow 12.47\%$
			4	70.89	—	—	—	33.54	36.60	35.07	$\downarrow 11.71\%$
			8	70.96	—	—	—	32.32	37.00	34.66	$\downarrow 12.73\%$
			16	70.89	—	—	—	32.32	36.60	34.46	$\downarrow 13.24\%$
	W4A8	DeltaQuaRot	32	71.72	—	—	—	31.71	36.20	33.95	$\downarrow 14.51\%$
			2	77.33	—	—	—	39.63	34.80	37.22	$\downarrow 6.29\%$
			4	76.27	—	—	—	39.63	34.80	37.22	$\downarrow 6.29\%$
			8	76.80	—	—	—	39.63	34.60	37.12	$\downarrow 6.55\%$
			16	76.27	—	—	—	39.63	35.60	37.62	$\downarrow 5.29\%$
		DeltaDuQuant	32	77.10	—	—	—	38.41	35.20	36.81	$\downarrow 7.33\%$
			2	58.45	—	—	—	22.56	31.20	26.88	$\downarrow 32.33\%$
			4	64.14	—	—	—	23.17	31.20	27.19	$\downarrow 31.56\%$
			8	68.46	—	—	—	29.27	33.80	31.53	$\downarrow 20.60\%$
			16	69.75	—	—	—	31.71	34.20	32.95	$\downarrow 17.03\%$
	W4A4	DeltaQuaRot	32	72.18	—	—	—	34.76	35.20	34.98	$\downarrow 11.93\%$
			2	61.11	—	—	—	20.12	24.40	22.26	$\downarrow 43.95\%$
			4	69.22	—	—	—	23.78	24.60	24.19	$\downarrow 39.09\%$
			8	73.39	—	—	—	32.32	27.40	29.86	$\downarrow 24.82\%$
			16	73.31	—	—	—	30.49	29.00	29.74	$\downarrow 25.11\%$
		DeltaDuQuant	32	75.21	—	—	—	37.80	33.00	35.40	$\downarrow 10.86\%$

Clipping Results (8 anchors)

Table 51: Effect of clipping on LLaDA-8B-Instruct accuracy results

Model	Setting	Method	Clip	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	–	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W4A8	DeltaQuaRot	8 anchors	70.96	–	–	–	32.32	37.00	34.66	↓ 12.73%
			0.9999	71.72	–	–	–	32.93	35.80	34.36	↓ 13.48%
			0.999	72.25	–	–	–	35.37	35.00	35.18	↓ 11.41%
			0.99	71.65	–	–	–	36.59	36.20	36.39	↓ 8.37%
		DeltaDuQuant	8 anchors	76.80	–	–	–	39.63	34.60	37.12	↓ 6.55%
			0.9999	77.10	–	–	–	40.24	33.20	36.72	↓ 7.54%
			0.999	76.35	–	–	–	37.80	35.00	36.40	↓ 8.35%
			0.99	76.27	–	–	–	37.20	34.80	36.00	↓ 9.36%
	W4A4	DeltaQuaRot	8 anchors	68.46	–	–	–	29.27	33.80	31.53	↓ 20.60%
			0.9999	68.31	–	–	–	26.83	30.60	28.71	↓ 27.70%
			0.999	68.54	–	–	–	29.88	34.60	32.24	↓ 18.83%
			0.99	69.60	–	–	–	31.71	34.00	32.85	↓ 17.28%
		DeltaDuQuant	8 anchors	73.39	–	–	–	32.32	27.40	29.86	↓ 24.82%
			0.9999	74.00	–	–	–	31.10	27.00	29.05	↓ 26.86%
			0.999	74.22	–	–	–	31.71	30.80	31.25	↓ 21.31%
			0.99	74.53	–	–	–	34.76	29.00	31.88	↓ 19.74%

Clipping Results (16 anchors)

Table 52:Effect of clipping on LLaDA-8B-Instruct accuracy results (16 anchors)

Model	Setting	Method	Clip	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	–	78.85	34.87	56.86	–	39.63	39.80	39.72	–
			16 anchors	70.89	–	–	–	32.32	36.60	34.46	↓13.24%
			0.9999	71.11	–	–	–	34.15	35.80	34.97	↓11.94%
			0.999	71.95	–	–	–	33.54	36.20	34.87	↓12.21%
	W4A8	DeltaQuaRot	0.99	72.40	–	–	–	36.59	36.80	36.69	↓7.62%
			16 anchors	76.27	–	–	–	39.63	35.60	37.62	↓5.29%
			0.9999	77.79	–	–	–	41.46	34.80	38.13	↓3.99%
			0.999	76.35	–	–	–	38.41	35.00	36.71	↓7.58%
			0.99	77.71	–	–	–	41.46	35.40	38.43	↓3.24%
	W4A4	DeltaQuaRot	16 anchors	69.75	–	–	–	31.71	34.20	32.95	↓17.03%
			0.9999	69.98	–	–	–	30.49	33.00	31.74	↓20.08%
			0.999	70.81	–	–	–	32.93	34.00	33.46	↓15.75%
			0.99	71.11	–	–	–	31.10	35.40	33.25	↓16.29%
	W4A4	DeltaDuQuant	16 anchors	73.31	–	–	–	30.49	29.00	29.74	↓25.11%
			0.9999	75.82	–	–	–	37.80	32.80	35.30	↓11.11%
			0.999	75.82	–	–	–	37.80	33.60	35.70	↓10.11%
			0.99	74.98	–	–	–	37.80	30.60	34.20	↓13.88%

Clipping Results (32 anchors)

Table 52: Effect of clipping on LLaDA-8B-Instruct accuracy results (32 anchors)

Model	Setting	Method	Clip	Math				Code			
				GSM8K	MATH	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	–	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W4A8	DeltaQuaRot	32 anchors	71.72	–	–	–	31.71	36.20	33.95	↓ 14.51%
			0.9999	70.66	–	–	–	32.93	36.00	34.46	↓ 13.23%
			0.999	71.57	–	–	–	31.71	36.40	34.05	↓ 14.26%
			0.99	71.87	–	–	–	36.59	36.60	36.59	↓ 7.87%
		DeltaDuQuant	32 anchors	77.10	–	–	–	38.41	35.20	36.81	↓ 7.33%
			0.9999	77.33	–	–	–	39.02	34.60	36.81	↓ 7.31%
			0.999	77.03	–	–	–	37.80	35.40	36.60	↓ 7.84%
			0.99	76.50	–	–	–	40.24	34.20	37.22	↓ 6.28%
	W4A4	DeltaQuaRot	32 anchors	72.18	–	–	–	34.76	35.20	34.98	↓ 11.93%
			0.9999	71.87	–	–	–	32.93	35.60	34.26	↓ 13.73%
			0.999	70.43	–	–	–	31.71	36.00	33.85	↓ 14.76%
			0.99	71.34	–	–	–	31.10	35.80	33.45	↓ 15.78%
		DeltaDuQuant	32 anchors	75.21	–	–	–	37.80	33.00	35.40	↓ 10.86%
			0.9999	76.65	–	–	–	35.37	32.40	33.88	↓ 14.69%
			0.999	76.04	–	–	–	42.68	34.00	38.34	↓ 3.46%
			0.99	76.04	–	–	–	37.20	32.20	34.70	↓ 12.64%

Combined Results

LLaDA-8B-Instruct accuracy: QuaRot vs. DuQuant rotation quantization, with and without delta-quant. For delta variants, *Config* reports the anchor count and clip ratio; the "1 anchor, no clip" rows are the original delta-quant setting, and the "best" rows report the top-performing sweep point (by Code Avg). $\Delta\%$ is the relative change of the group average w.r.t. the FP16 base.

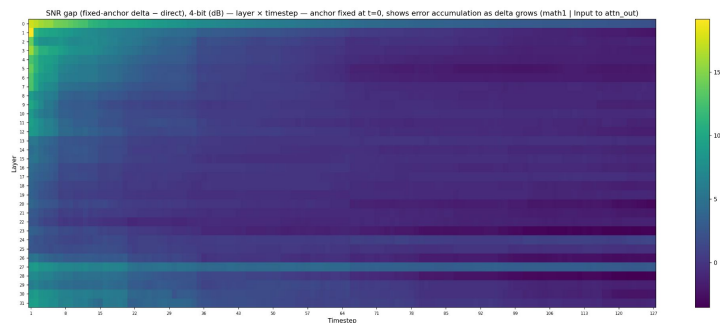
Model	Setting	Method	Config	Math				Code			
				GSM8K	MATH500	Avg	$\Delta\%$	HumanEval	MBPP	Avg	$\Delta\%$
LLaDA-8B-Instruct	FP16	Base	–	78.85	34.87	56.86	–	39.63	39.80	39.72	–
	W4A8	QuaRot	–	71.65	–	–	–	35.37	36.00	35.68	↓10.16%
		DeltaQuaRot	1 anchor, no clip	71.80	–	–	–	32.93	36.60	34.76	↓12.47%
		DeltaQuaRot	16 anchors, clip 0.99	72.40	–	–	–	36.59	36.80	36.69	↓7.62%
		DuQuant	–	77.63	–	–	–	39.02	34.60	36.81	↓7.31%
		DeltaDuQuant	1 anchor, no clip	77.33	–	–	–	39.63	34.80	37.22	↓6.29%
		DeltaDuQuant	16 anchors, clip 0.99	77.71	–	–	–	41.46	35.40	38.43	↓3.24%
	W4A4	QuaRot	–	67.55	–	–	–	26.83	32.00	29.41	↓25.93%
		DeltaQuaRot	1 anchor, no clip	49.36	–	–	–	21.34	31.00	26.17	↓34.11%
		DeltaQuaRot	32 anchors, no clip	72.18	–	–	–	34.76	35.20	34.98	↓11.93%
		DuQuant	–	75.74	–	–	–	35.37	29.80	32.58	↓17.96%
		DeltaDuQuant	1 anchor, no clip	45.49	–	–	–	19.51	23.60	21.56	↓45.73%
		DeltaDuQuant	32 anchors, clip 0.999	76.04	–	–	–	42.68	34.00	38.34	↓3.46%

FreeAct

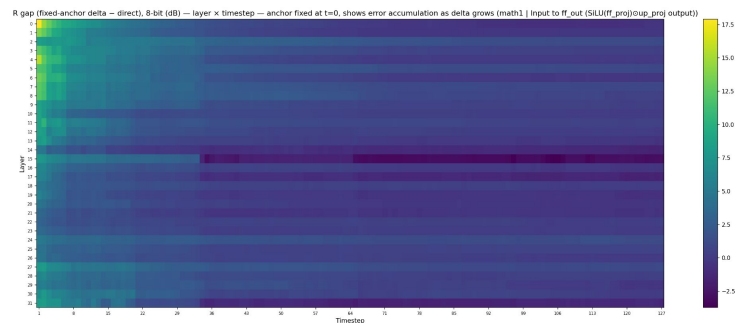
	dLLMs			
	HumanEval	GSM8K	Math500	Avg.
	<i>LLaDA</i>			
16-bit Baseline	43.90	76.80	38.40	53.03
RTN (W8A8)	39.02	77.56	37.00	51.19
RTN (W4A4)	00.00	00.00	00.14	00.05
SmoothQuant	00.00	00.00	00.00	00.00
QuaRot	28.66	65.81	24.20	39.56
FlatQuant	39.63	74.37	33.60	49.20
FreeAct (ours)	42.68	75.74	34.60	51.01

SNR Gap - ff_out_in activations (1 fixed anchor)

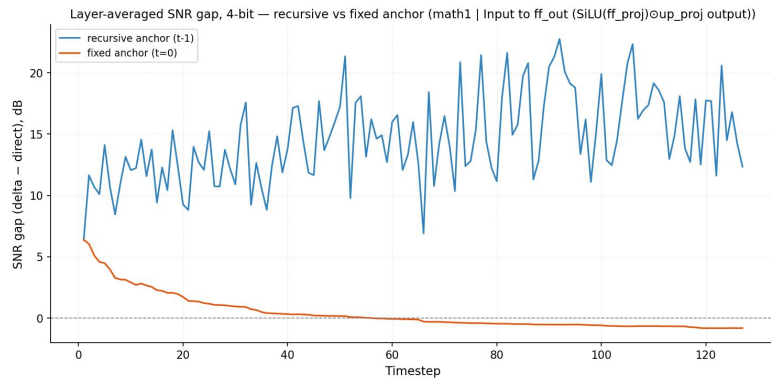
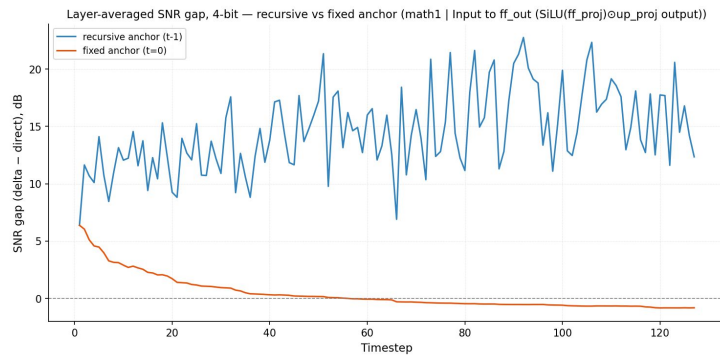
4-bit SNR Gap



8-bit SNR Gap



Recursive vs Fixed Anchor SNR Gaps



Plan for Next Week

- Run experiments using non-uniform reanchoring methods
- Implement Delta Quant on FlatQuant
- Report the results for Math500